

Exploring the Dichotomy of Nature and Tech Through an Interactive Media Assignment

This document outlines an interactive media assignment.

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ABSTRACT

In this submission I present a mixed media assignment that asks students to challenge the dichotomy of nature and technology. For this assignment, students use animation, 3D modeling, and physical making to create an interactive piece that embodies the aesthetics of nature through visuals and sound or visuals and touch. The primary goal of this assignment is to strengthen a connectedness to nature through creativity, while also challenging the students by encouraging mixing media in creative ways. The current description of this assignment is open and flexible; however, it can also be focused on a specific discipline (e.g. animation or game development) or scaled by adjusting the scope of the final deliverable. This paper will give a brief overview of the assignment including the prompt, process, materials, and examples of student work.

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Media arts.

KEYWORDS

Immersive media, animation, interactive art, mixed media

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1 INTRODUCTION

Nature and Technology are often thought of as two dichotomous, or at the very least, conflicting ideas. However, since technology is such an integral part of our lives, what if, combined with art, it became part of the solution? This assignment asks students to challenge that dichotomy by leveraging technology to create a mixed media, interactive piece which embodies visual and auditory, or visual and tactile aesthetics from nature. The primary goal is to explore experiences in natural environments and translate those to interactive art pieces to not only foster an increased connectedness to nature in the students, but others who experience their work.

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Table 1: Metadata in tabular form.

Summary	Mixed media, interactive piece based on nature and technology
Learning outcomes	Combine media in interesting ways, create an interactive art piece that addresses the prompt
Classification	Animation, modeling, graphics and interfaces
Audience	Sophomore and higher
Dependencies	Digital illustration, basic 3D modeling, intro game engine
Prerequisites	None
Strengths	Scale-able and applicable to different disciplines, encourages students to think about their work in a larger context, creative use of media-making techniques
Weaknesses	Open-ended could lead to decision paralysis
Variants	Unlimited or could be more focused on specific disciplines
Assessment	Proper adherence to theme, creative media mixing, overall quality

In addition, the mixing of different forms of physical and digital media challenges and strengthens creativity in their media making skills.

To begin students are given a short reading list to prepare them for addressing the prompt. They then work in teams to ideate, prototype, and create the final mixed media piece. Teams are organized by each student's diverse skillset to encourage creative media mixing. For example, one team combined clay sculpting with 3D printed models to create a tide pool diorama which sits over a screen displaying interactive 2D animated fish. Other successful examples include an immersive cave viewed through a physical foreground and digital background in Unity, and a deep-sea experience with 2D animations projected in 360 degrees and physical glow in the dark sculptures.

2 METADATA

3 MATERIALS

The materials, software, and tools vary greatly depending on the design of the piece. Ultimately, they can be broken down into three

categories: physical materials, software, and digital platforms. For physical materials, assets can be created with found materials (e.g. cardboard, paper, wire), clay or other sculpting materials, and even 3D printing. Additionally, supplemental materials such as paint, glue, and tools to manipulate physical media should be considered.

Software and digital platform wise, students are free to use what best fits their design. Typically, I recommend Procreate and After Effects for 2D assets and motion, Blender for 3D modeling and 3D printing preparation, and real-time game engines such as Unity or Unreal for their final digital platform. Students are also encouraged to consider additional technology such as screens, projectors, and VR headsets in the implementation of their final pieces.

4 CONCLUSION

Overall, this assignment presents students with an opportunity to think about their work in a broader context while simultaneously challenging their ability to create media in creative ways. Students

show a lot of curiosity and playful experimentation with nature aesthetics which feeds into a greater connection the natural world. Completion of this assignment demonstrates success in fostering creativity, encouraging the exploration of nature aesthetics, and pushing students to create interactive pieces in interesting ways.

Future implementations of this assignment would experiment with more limitations and constraints such as requiring specific media, themes, or tools and software. While some students enjoyed the freedom, many struggled with how open and flexible the assignment was. Working within constraints is not only a common way to foster creativity, but it could also help steer the assignment toward specific outcomes and challenge students in ways they may not push themselves to otherwise.

Ultimately the complexity, creativity, and quality of the resulting pieces illustrate the assignment's success. Additionally, students and those who experienced their work appreciated the theme of nature versus technology and described the pieces in ways they might describe experiences in nature: with wonder and curiosity.